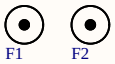
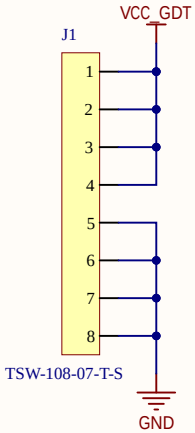
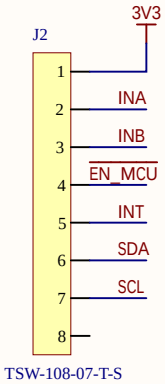
Fiducials

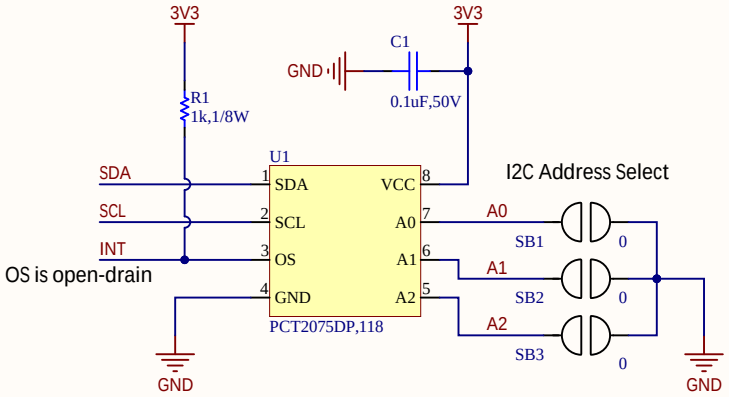


4.5V <= VCC_GDT <= 26V

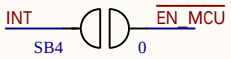




Temperature Sensor

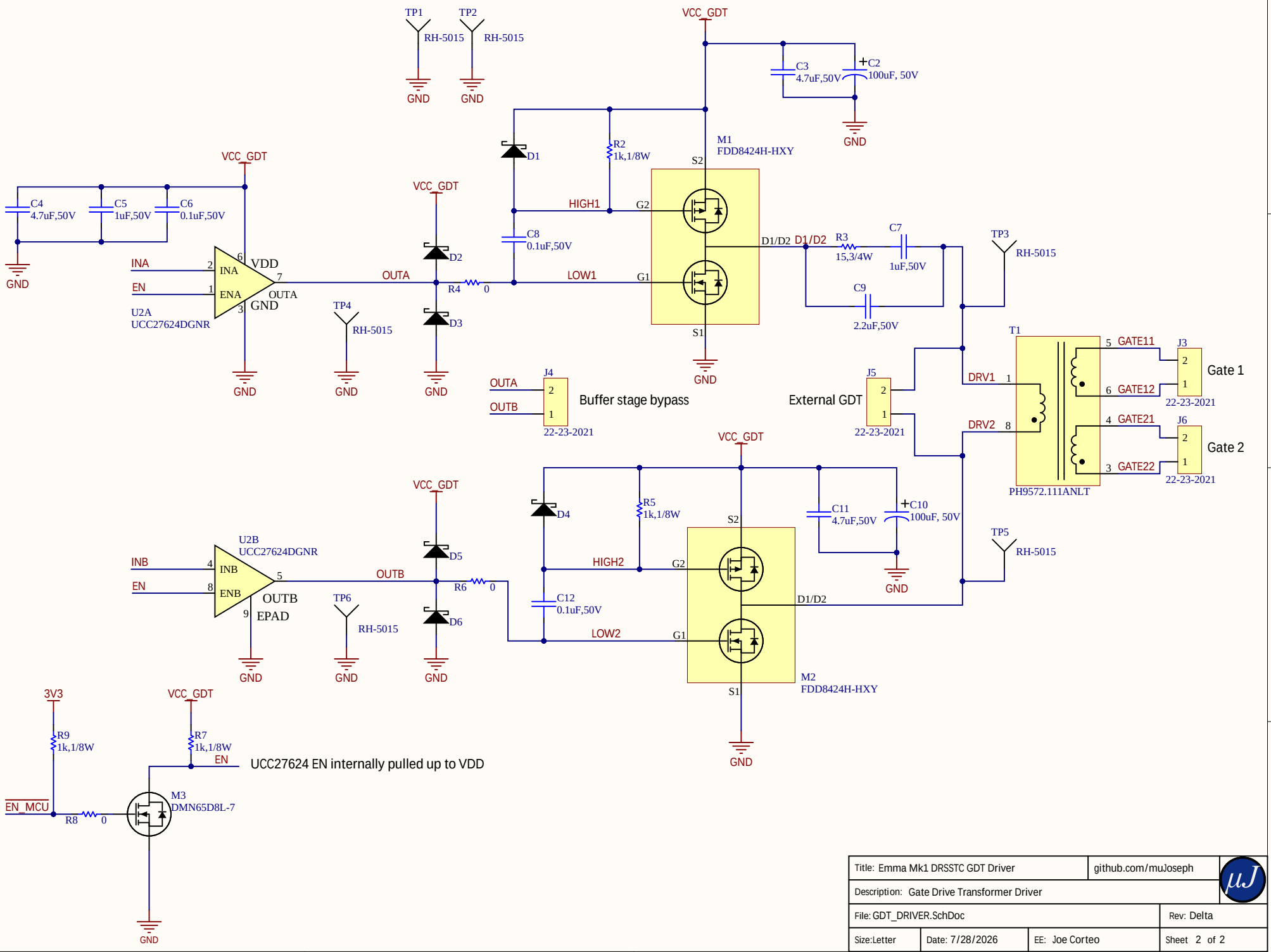


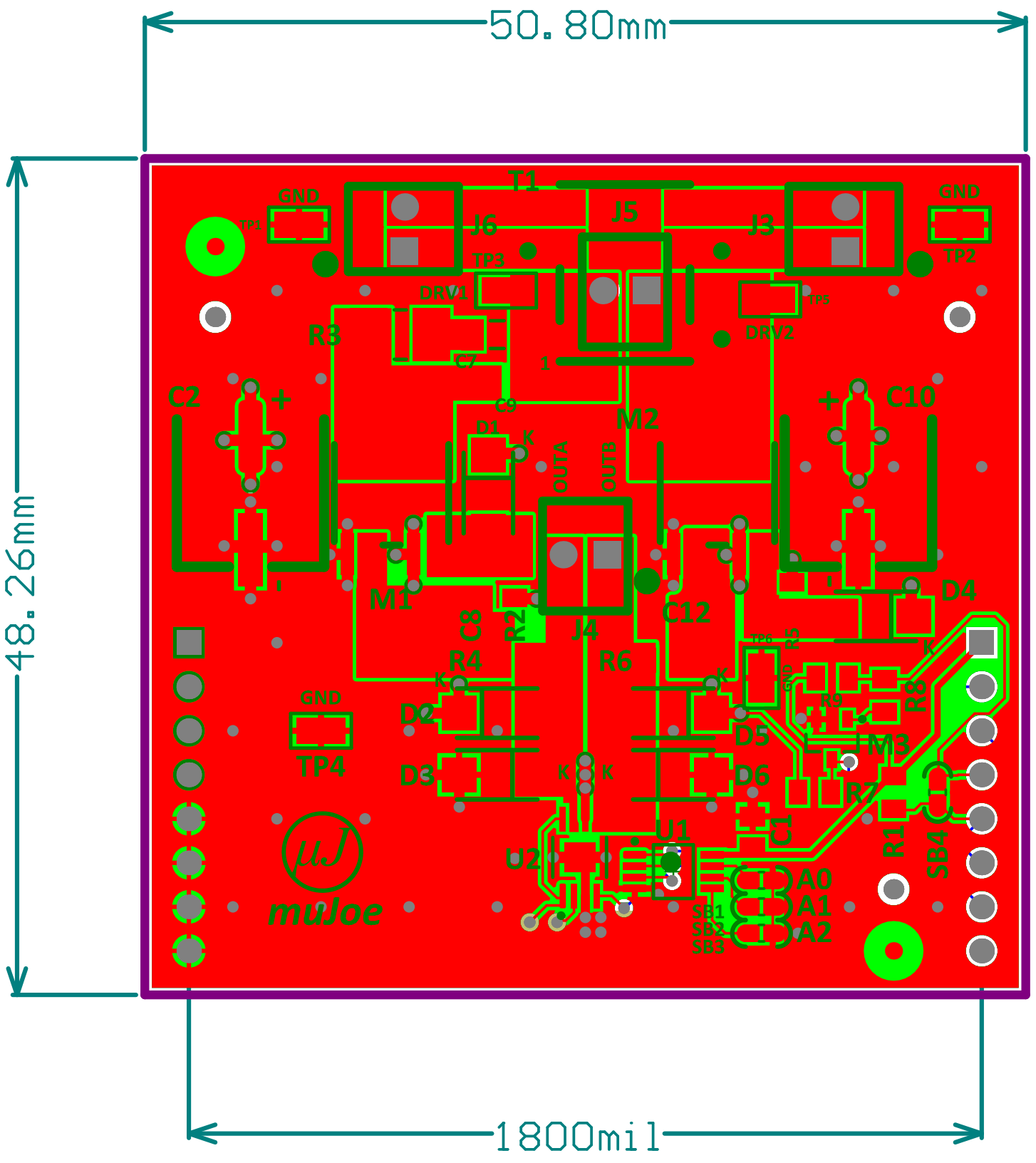
Auto Overtemp Shutdown



NOTE: If SB4 is 0, R9 must be DNP

Title: Emma Mk1 DRSSTC GDT Driver		github.com/muJoseph	
Description: Mechanical and Temperature Sensor			
File: MECH_TEMP.SchDoc			Rev: Delta
Size:Letter	Date: 7/28/2026	EE: Joe Corteo	Sheet 1 of 2


TP1
TP2
RH-5015
RH-5015
GND
GND
VCC_GDT
C3
4.7uF,50V
C2
100uF, 50V
GND
D1
R2
1k,1/8W
M1
FDD8424H-HXY
S2
VCC_GDT
C4
4.7uF,50V
C5
1uF,50V
C6
0.1uF,50V
HIGH1
G2
C8
0.1uF,50V
D1/D2
D1/D2
R3
15,3/4W
C7
1uF,50V
TP3
RH-5015
INA
EN
U2A
UCC27624DGNR
VDD
INA
ENA
GND
OUTA
OUTA
D2
R4
0
LOW1
G1
C9
2.2uF,50V
TP4
RH-5015
D3
S1
GND
GND
GND
GND
OUTA
OUTB
J4
22-23-2021
Buffer stage bypass
External GDT
J5
22-23-2021
T1
DRV1
DRV2
5
GATE11
J3
Gate 1
6
GATE12
22-23-2021
4
GATE21
J6
Gate 2
3
GATE22
22-23-2021
PH9572.111ANLT
VCC_GDT
D4
R5
1k,1/8W
C11
4.7uF,50V
C10
100uF, 50V
S2
TP5
RH-5015
INB
EN
U2B
UCC27624DGNR
INB
ENB
OUTB
EPAD
OUTB
D5
R6
0
HIGH2
G2
C12
0.1uF,50V
D1/D2
TP6
RH-5015
D6
LOW2
G1
GND
GND
GND
GND
S1
M2
FDD8424H-HXY
3V3
VCC_GDT
R9
1k,1/8W
R7
1k,1/8W
EN
UCC27624 EN internally pulled up to VDD
EN_MCU
R8
0
M3
DMN65D8L-7
GND
Title: Emma Mk1 DRSSTC GDT Driver
github.com/muJoseph
Description: Gate Drive Transformer Driver
File: GDT_DRIVER.SchDoc
Rev: Delta
Size:Letter
Date: 7/28/2026
EE: Joe Corteo
Sheet 2 of 2


50.80mm
48.26mm
1800mil
T1
J6
J5
J3
GND
TP1
TP2
TP3
TP4
TP5
TP6
DRV1
DRV2
R3
C7
C9
C2
C10
M2
D1
OUTA
OUTB
M1
C8
R2
J4
C12
R5
D4
R4
R6
D2
D5
R9
R8
M3
D3
D6
R7
U1
U2
R1
SB4
A0
A1
A2
SB1
SB2
SB3
muJoe